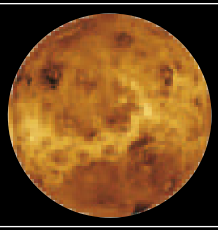
AN OCEAN PLANET

Planet Earth is really a planet of oceans. They cover more than two-thirds of its surface, to an average depth of 2.4 miles (3.8 km), making them the Earth's dominant environment. It is the only planet in the solar system that has liquid oceans, and it is no coincidence that it is the only planet known to support life. Water is vital to all forms of life, and it is likely that life on Earth began in the oceans. Oceans also make life on land possible, so without them humans could not exist.

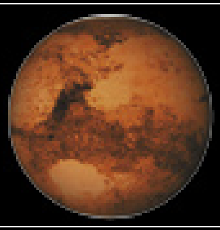
▲ IDEAL SITUATION
Earth's oceans exist due to a combination of lucky circumstances. The planet's distance from the Sun creates ideal temperatures that allow water to exist as a liquid. Its size and gravity enable it to retain an atmosphere of air and water vapor. This acts as insulation, and stops the oceans from boiling away or freezing solid.

OTHER WORLDS

MERCURY
The closest planet to the Sun, Mercury is much smaller than Earth. It does not have enough gravity to retain an insulating atmosphere, so its temperature fluctuates from -292°F (-180°C) to 806°F (430°C) and it is just a barren rocky sphere.



VENUS
Nearly the same size as Earth, Venus has enough gravity to retain a thick atmosphere. But this is mostly carbon dioxide, which traps the Sun's heat and creates searing surface temperatures of up to 930°F (500°C).



MARS
Too small to have more than a thin atmosphere, Mars is also colder than Earth because it is farther away from the Sun. Mars does have some water, but this is frozen into ice at its poles and beneath its surface, so the water does not form oceans.

EUROPA
One of the moons of Jupiter, Europa is rocky with a surface of solid ice, but it is possible that there is liquid water beneath the ice. If so, it might be the only other planet in the solar system that has extensive water, and a chance of supporting life.

▲ DEEP BLUE
The oceans have a total volume of about 319 million cubic miles (1,330 million cubic km). This is about a thousand times the volume of land above sea level. Living things can occupy all this space, and not just live on or near the surface as they do on land. So the total living space in the oceans is colossal, making them the most important habitat for life on Earth.